

Code-Layout, Readability and Reusability

Date	5 July 2026
Team ID	–
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design enhances maintainability, supports collaboration, and reduces overall project complexity.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Fulfilled (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform indentation used throughout the code	Yes	PEP8 followed throughout the project.
2	Proper File Structure	Files are logically organized by feature/module	Yes	Organized using modular folder structure.
3	Meaningful Variable Names	Variables and functions use clear, descriptive names	Yes	Naming follows lower_snake_case nomenclature.
4	Function / Method Names	Functions/methods carry relevant and descriptive names	Yes	Names represent the purpose of the function.
5	Code Comments	Inline and block comments explain logic where necessary	Partial	Major modules commented; more inline comments can be added.
6	Code Modularity	Code is split into reusable modules and components	Yes	Services and utilities are well modularized.
7	No Hardcoded Code	Sensitive or reusable data is not hardcoded	Yes	Environment variables used for secrets and config.
8	Error Handling	Exceptions are captured and handled gracefully	Yes	Try-except blocks and custom error messages implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Audio Upload & Validation	Python, FastAPI	Used in upload API and admin tools	High

2	Speech-to-Text Service	Python, Whisper API	Used in transcription pipeline and re-evaluation	High
3	Text Preprocessing Utilities	Python	Used in NLP analysis and training pipeline	High
4	NLP Analysis Pipeline	Python, scikit-learn, Transformers	Used in scoring and concept matching	High
5	Scoring & Feedback Engine	Python	Used in feedback and report generation	Medium

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	4.5	Well-structured project with consistent layout and modular organization.
Readability	4	Code is easy to read with meaningful names and good documentation.
Reusability	4.5	High reuse of components (STT, NLP pipeline, utils). Easily extendable.
Documentation / Comments	4	Good overall, but some functions can use more detailed docstrings.
Overall Score	4.2 / 5	High quality codebase with strong modularity and maintainability.